

## Power Cube Family



Integrated Controller for Managing  
Predefined and Programmable Heating Cycles

# Master Controller V3+

## Operator Manual

| Document           | Date       | Software |
|--------------------|------------|----------|
| FI040K0021v1000hUK | 04/02/2021 | MCPC35xx |



Read this manual carefully before operating or carrying out maintenance on the device. Keep the manual in a safe place for future reference, and in perfect condition. This manual must accompany the device described therein in the case of change of ownership, and until the device is broken up.

## SYMBOLS



The equipment is marked with this symbol wherever the user should refer to this manual in order to avoid possible damage. The same symbol appears in the manual at points where warnings or particularly important instructions, essential for safe, correct operation of the device, are given.



The equipment is marked with this symbol in the areas where there is dangerous voltage. Only trained maintenance personnel should carry out work in these areas. The same symbol appears in the manual at points where warnings essential for safe, are given.



This symbol appears in the manual at points where suggestions, additional information or other relevant notes are given.

## REVISION TABLE

| Rev. | Date       | Author   | Reference | Description    |
|------|------------|----------|-----------|----------------|
| 1000 | 2021/02/04 | DTP – SP | -         | First Emission |

## Warranty terms

The warranty on all CEIA products, extended to the period agreed with the Sales Department, is applicable to goods supplied from our factory, and for every constituent part thereof, with the exception of the batteries. Any form of tampering with the device, and in particular opening its container, is strictly forbidden and will invalidate the warranty.

### Customer Satisfaction Report

C.E.I.A. objective is the maximum quality of products and services offered to the customer. Any comments and suggestions useful for achieving this objective will be highly appreciated and can be sent to us by compiling and returning the form available at:

<http://www.ceia-power.com/satisfaction>

## Manufacturer

**CEIA S.p.A.**  
**Costruzioni Elettroniche Industriali Automatismi**  
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Read this manual carefully before installing or operating the device and before carrying out maintenance operations.

## 1 SAFETY INSTRUCTIONS – WARNINGS

### 1.1 Warnings

#### 1.1.1 General warnings

- Follow the instructions contained in this manual for all operations relating to Operator of the device.

CEIA cannot be held responsible for any damage resulting from procedures which are not expressly indicated in this manual, or from any lack of attention, either partial or total, of the procedures described therein.

- Whenever there is any suggestion that the level of protection has been reduced, the device should be taken out of service and secured against any possibility of unintentional use, and authorised service technicians should be called.

The level of protection is considered to have been reduced when:

- *the device shows visible signs of wear and tear, especially on parts that ensures the device protections (boxes, gaskets, glands, fixing screws, etc.);*
- *the ground connections are not compliant;*
- *the device has suffered mechanical or electrical stress (shocks, bumps, etc.);*
- *the device does not operate correctly, respect on how described in this manual;*
- *the device has been stored for long periods in sub-optimal conditions;*
- *the device has suffered severe stress during transport;*
- *the inside of the device has come into contact with liquids.*
- Handle the device with care and without excessive force during operation.

#### 1.1.2 Use warnings

- To avoid damage due to lightning, disconnect the power-supply cable during storms.
- Never cut off the safety features of the device.
- This device contains electrical and electronic components, and may therefore be susceptible to fire. Do not install in explosive atmosphere or in contact with inflammable material, Do not use water or foam in the case of fire when the device is powered up.

#### 1.1.3 Maintenance warnings

- Before any maintenance, cleaning or movement, the device must first be disconnected from all sources of power.
- Always remove the plug by hand when disconnecting the power supply cable, never by pulling on the cable.
- Do not wash the device with liquid detergents or chemical substances. Any cleaning should be done using a slightly-damp, non-abrasive cloth.
- Read the chapter on "Maintenance" carefully before calling the service centre. Whatever the problem, only specialised service personnel authorised to work with CEIA equipment should be called.
- Any damaged parts of the device should be replaced with original components only.
- Any maintenance or repair of the device while open and energised should be avoided, and in any case should only be carried out by trained personnel who are fully aware of the risks which the operation entails, following the instructions given in the Maintenance section.

## **2 DESCRIPTION**

### **2.1 General description**

#### **2.1.1 Description**

The **Master Controller V3+** is a computerized industrial control unit housed in a robust stainless steel container, designed for automatic and semi-automatic management of heating cycles carried out using CEIA Power Cube Induction Generators.

The Controller comes with a set of predefined programs. All operating parameters for each phase of the heating cycle can be programmed within a wide range of values. This allows maximum flexibility in setting the heating and cooling times, the power supplied by the induction generators, the quantity of solder wire and the operating temperatures for the piece being worked on.

Together with the optical temperature sensors (SH/SLE series), and the soldering wire feeder (PW3-WF series), the **CEIA Master Controller V3+** allows the creation of efficient induction brazing systems and achievement of completely repeatable heating processes and finished products.

An RS-232 serial interface, or an optional Field Bus interface, allows connection to a PLC or other external logic for remote loading and running of operating programs.

The photo-coupler-type static inputs have programmable P/N logic, and together with the static outputs are subject to a programmable auxiliary voltage of 12/24 V.

### **2.2 Correct and Forbidden use of the device**

#### **2.2.1 Correct use of the device**

The Master Controller V3+ is an electronic device for the automatic and semi-automatic management of heating cycles executed by induction generators CEIA PowerCube.

#### **2.2.2 Forbidden use of the device**

The electrical plug must only be inserted into a socket containing a grounding contact. Any interruption in the protective conductor either inside or outside the device, or the detaching of the protective grounding terminal, renders the device dangerous. Intentional interruption is forbidden.

## 2.3 Components

### 2.3.1 Control unit

The device includes a power supply stage and a programmable command section. The apparatus is guided by a microcomputer, which, thanks to its special programming, has the capacity to simultaneously control the entire soldering process of two soldering/heating points (ignition; regulation of the temperature, power provided, period of activity, synchronisation of all the phases, etc.).

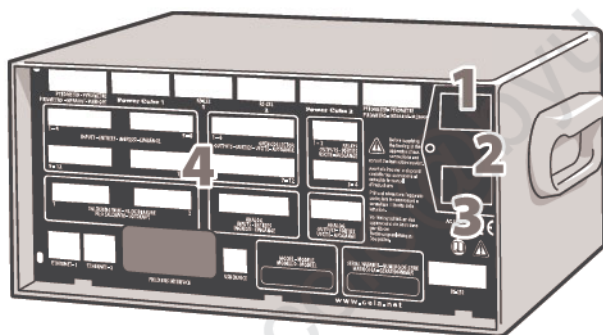
The Master Controller monitors the correct functioning of the machine by means of a self-diagnosis device and signals, by display, any necessary interventions on the part of the operator.

On the front panel are:



|          |                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------|
| <b>A</b> | Graphic display, indicating the messages referred to the functioning, programming and self-diagnosis of the system. |
| <b>B</b> | Control keypad composed by the following function keys                                                              |
| <b>C</b> | Flash Card reader/writer to load and save different working parameters sets.                                        |
| <b>D</b> | USB port (not used).                                                                                                |

On the back panel of the generator are the following:



|          |                                          |
|----------|------------------------------------------|
| <b>1</b> | Power supply plug                        |
| <b>2</b> | Power supply fuse (1A, 250V ; time lag ) |
| <b>3</b> | ON/OFF switch                            |
| <b>4</b> | Connections                              |

## 2.4 Safety features

The **Master Controller V3+** is fitted with a list of safety features described below, in order to warrant the proper protection level, during the correct use.

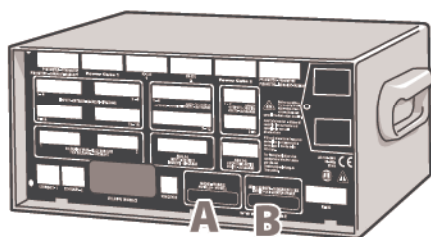
|                              |                                                                                                                                                                                    |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IP Protection degree</b>  | IP20                                                                                                                                                                               |
| <b>Self-diagnosis system</b> | An internal self-diagnosis system checks constantly the functionality of the system. In case of fault a message is shown on the display of the control panel and of the generator. |
| <b>Password</b>              | Access to parameters and programs of the Controller is protected by passwords. USER and SUPER USER access level are available.                                                     |



## 2.5 Options

| Option                                                                                                                                                                          | Order code                                                                                                  | Symbol |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------|
| <b>Master Controller V3+</b><br>Standard version                                                                                                                                | <b>MASTER-C-V3+</b>                                                                                         | -      |
| <b>Wire Feeder option</b><br>Management up to two independent wire feeders<br>(one for each soldering point)<br><i>Wire feeder not included.</i>                                | <b>V3+/WF</b>                                                                                               |        |
| <b>Thermal Profile option</b><br>Thermal Profile management                                                                                                                     | <b>V3+/TP</b>                                                                                               |        |
| <b>IXC option</b><br>Web Server management                                                                                                                                      | <b>V3+/IXC</b>                                                                                              |        |
| <b>Field Bus interface</b>                                                                                                                                                      | <b>V3+/FB-ETHERCAT<br/>V3+/FB-ETHERNET/IP<br/>V3+/FB-MODBUS-1<br/>V3+/FB-PROFIBUS-1<br/>V3+/FB-PROFINET</b> |        |
| <b>Tin Soldering option</b><br>Version optimized for Tin Soldering.<br><i>Requires the -WF option</i>                                                                           | <b>V3+/TS</b>                                                                                               |        |
| <b>DIAS option</b><br>Management of DIAS thermocamera.<br><i>Requires the -IXC option</i>                                                                                       | <b>V3+/DIAS</b>                                                                                             |        |
| <b>Stand-Alone option</b><br>Management up to two optical sensors and two wire feeders, without<br>the connection with a CEIA generator.<br><i>Includes -WF and -TS options</i> | <b>V3+/STANDALONE</b>                                                                                       |        |
| <b>Sensors Only option</b><br>Management up to two CEIA SLE pyrometers, without the connection<br>with a CEIA generator.                                                        | <b>V3+/SO</b>                                                                                               |        |
| <b>Analog Sensors option</b><br>Temperature reading by external analog sensors<br>(0-10V, 4-20mA, 0-20mA).                                                                      | <b>V3+/ANALOG2S</b>                                                                                         |        |
| <b>Synchronized Sections option</b><br>Allows the simultaneous activation of two CEIA generators on the<br>same work piece.<br><i>Requires the -TP option</i>                   | <b>V3+/SYNSEC</b>                                                                                           |        |

### 2.5.1 Device identification



**A:** Indication of the model and  
any installed option



**B:** Serial Number



### 3 USE

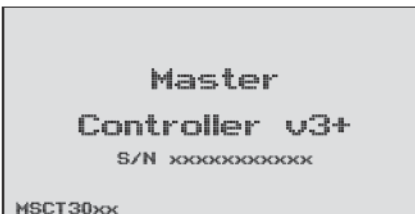


Before starting any operation on the unit, read carefully the **Safety instructions - Warnings** section of this manual.

#### 3.1 Power ON

Activate the main switch located on the back of the Controller.

Powering on the Master Controller, the display shows, for a few seconds, a screenshot indicating the Controller model, the serial number and the software version, as shown beside.



#### 3.2 Indicators

##### 3.2.1 Optical indicators

All optical indications given by the Master Controller V3+ are shown on the display.  
Beside a work screenshot example.

```
1> T= 700C  S=====L
    TIME=----s  POW.=99%
2> T= 700C  S=====L
    TIME=----s  POW.=99%
```

##### 3.2.2 Acoustical indicators

The Master Controller V3+ utters only a short acoustical signal when a self-diagnosis message is displayed and two short acoustical signals when the fault is solved and is ready to operate again.

#### 3.3 Controls

Programming and Management are executed through the built-in keypad described below:

| KEYS                                                                                | FUNCTIONS                                               |
|-------------------------------------------------------------------------------------|---------------------------------------------------------|
|  | Access to and exit from Programming phase               |
|                                                                                     | Return to the previous menu                             |
|  | Scrolling of the instructions sequence                  |
|                                                                                     | Selection of the parameter to be modified               |
|                                                                                     | Parameter value modification                            |
|  | Selection of the submenu                                |
|                                                                                     | Value confirmation                                      |
|  | To check the status of the generator                    |
|  | To access the quick menu programmable by the Superuser. |



### 3.4 Use of the device

#### 3.4.1 Start-up

Check that all the devices needed for the heating operation are placed correctly, so as to let the workstation operative.

Supply all system devices with the right voltage and activate all workstation components.

Activate the main switch located on the back of the Controller.

#### 3.4.2 Set-up

The display of the Master Controller is divided, in this case, in two sections: the upper row shows the status of the soldering point 1, the lower row the status of the point 2 (see picture beside). Even in the case of a self-diagnosis message, this will be displayed in the row related to the heating point that generated the message.

```

1> T= 700C  PID S====#===L
WIRE= 50,0mm POW.=99%
                2.6s
2> T= 800C  2C S====#===L
WIRE= 50,0mm POW.=99%
        6+MEM006        2.6s
    
```

For each heating point, a bar-graph indication is given to represent the inductor dimensioning: if the cursor is between the two characters "S" and "L" the dimensions are correct.

If the cursor moves to the left until the "S" character (small) starts to blink, the inductor is too small.

If the cursor moves to the right with the "L" character (large) blinking, the inductor is too big.

Beside the temperature indication, a small symbol indicates if a thermocouple (PID) or a two-colours pyrometer (2C) is connected. This latter indication appears only if the pyrometer is operating in two-colours mode, over 600°C, because for temperatures lower than that, the pyrometer uses only one sensor.

In the example above, the section 1 is connected to a thermocouple and in the section 2, an SH2C/SLE two-colours pyrometer is operative.

#### 3.4.3 Normal use

The Controller will operate with the parameters set during the installation. If needed, load a new set of parameters from the memory.



Each time the Master Controller is turned on, it automatically sets itself using the last parameters in use.

Place the pieces to be heated near the heating inductor, taking care not to be in contact with it. Activate the heating cycle, and carry out the required operations.

### 3.4.4 Status screenshots

From the work screenshot, pressing the STATUS key, the generator status screen is displayed. Press again STATUS and the work screenshot will be displayed again.

|           |                                                           |                                                                                                                                                           |
|-----------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1>        | Current diagnosis status on point 1                       | <div>STATUS</div> <div>1&gt; OK<br/>PREVIOUS: A3</div> <div>2&gt; OK<br/>PREVIOUS: A3</div> <div>MST63.028.0231--<br/>S/N: 213040039999<br/>OPT: WF</div> |
| 2>        | Current diagnosis status on point 2                       |                                                                                                                                                           |
| PREVIOUS: | Diagnosis message previously shown on corresponding point |                                                                                                                                                           |
|           | Software version currently installed                      |                                                                                                                                                           |
| S/N       | Serial number                                             |                                                                                                                                                           |
| OPT:      | Installed options                                         |                                                                                                                                                           |

### 3.4.5 Manual advancing of the soldering wire

Whenever necessary, it is possible to activate the feeding motor of the soldering wire forwards and backwards, in order to set up the work station or the unblock a possible jamming of the wire. This function is accessible through the MANUAL programming menu (see paragraph 3.6.3).

### 3.4.6 Stop

When work is finished disconnect the supplies.

## 3.5 Program Cards Management

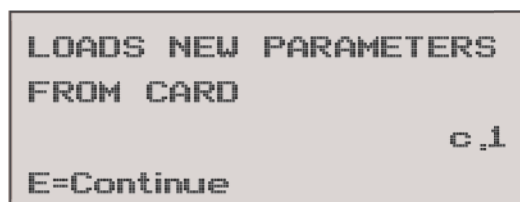
The **Master Controller** is provided with a Program Card reader, where it is possible to load and save until 100 different sets of working parameters.

### 3.5.1 Loading a set of parameters from the Program card



Loading a new set of parameters from the Program Card, the actual set of parameters will be overwritten, without any possibility to restore them.

To insert the Program Card, hold it with the golden contacts down and towards the Controller, and insert it until the display shows the following message:.



Pressing **E** all parameters stored in the card, at the selected memory cell, will be restored in the Controller.



### 3.5.2 Loading a set of parameters from the internal memory

This function is accessible through the MEMORY programming menu (see paragraph 3.6.2).

### 3.6 Programming



The programming of the Master Controller parameters can only be carried out by specialized technical personnel who are authorised to access the programming section.

The access to programming parameters, if set by the SUPERUSER, is protected by a password.

Pressing **P** the request of a password insertion is shown, before being able to access Programming.

Pressing **E** will allow to jump the password request and to access as USER to the programming parameters

If the password is correct, the programming phase is started and on the display is shown the main menu. The selected menu item is indicated with uppercase characters and the presence, on the left, of an arrow. On the bottom right corner of the display a letter shows the current access level: U=User S=Super User

**Password**

-----  
E: login as user

heating > **MODE**  
wire **memory**  
gas **manual**

S

- the main menu contains submenus that allow access to the operating parameters; scroll through the menus using the **UP** (increase) and **DOWN** (decrease) keys, and access the submenus by pressing the **E** key.
- menu voices in grey are not accessible at USER level.
- the parameters are activated with the **E** key.
- the parameter values are modified using the **UP** (increase) and **DOWN** (decrease) keys.
- modifications are confirmed with the **E** key.
- return to the main menu and exit from programming with the **P** key.

There follows an example of the programming operation with the messages that appear on the display and the keys pressed:

| Keys pressed | Displayed message                                                | Meaning                                                                                                                                                                            |
|--------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | heating > <b>MODE</b><br>wire <b>memory</b><br>gas <b>manual</b> | On the main menu, the selected submenu is the one written in capital letters and marked by the arrow (>)                                                                           |
|              | <b>LANGUAGE</b><br>> <b>ENG</b> <                                | Access to the chosen menu is made by pressing <b>E</b> key. First function is shown, with its actual value marked by two arrows                                                    |
|              | <b>CHANGE PASSWORD</b>                                           | Select the next function by pressing <b>DOWN</b> key                                                                                                                               |
|              | <b>NEW PASSWORD</b><br>-----                                     | Press <b>E</b> key to select the <b>Change password</b> function.                                                                                                                  |
|              | <b>NEW PASSWORD</b><br>* * * 3---                                | Scroll each character using the <b>UP</b> and <b>DOWN</b> keys, then press the <b>E</b> key to confirm and switch to the next character.                                           |
| o            | <b>CHANGE PASSWORD</b>                                           | Pressing the <b>E</b> key after the last character, confirm the new value. Pressing the <b>P</b> key in any moment, cancel the operation and send you back to the previous screen. |
|              | heating > <b>MODE</b><br>wire <b>memory</b><br>gas <b>manual</b> | Return to the preceding menu by pressing the <b>P</b> key and exit from the programming section by pressing the key again                                                          |

### 3.6.1 MODE menu

| Parameter                                   | Remote    | Possible settings               | Standard setting | Options |
|---------------------------------------------|-----------|---------------------------------|------------------|---------|
| <b>LANGUAGE</b>                             | <b>LG</b> | ENG, POL, SPA, DEU,<br>FRA, ITA | ENG              |         |
| Selects the language of displayed messages. |           |                                 |                  |         |

| Parameter                                                                                                                                         | Remote | Possible settings         | Standard setting | Options |
|---------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------|------------------|---------|
| <b>CHANGE PASSWORD</b>                                                                                                                            | -      | 6 alphanumeric characters | -                |         |
| Allows to insert a password for the current programming access level, or to change the existing one.<br>To remove the password, insert 6 spaces.. |        |                           |                  |         |

### 3.6.2 MEMORY menu

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                | Remote    | Possible settings          | Standard setting | Options |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------|------------------|---------|
| <b>LOAD NEW PARAMETERS FROM THE CELL</b>                                                                                                                                                                                                                                                                                                                                                                                                 | <b>LD</b> | from 1 to 100<br>0 to undo | -                |         |
| This operation selects the various parameters, previously stored, activating them for the next work cycle. when confirming the loading, all the parameters included unto the Controller work memory are replaced. Such parameters are stored into memory cell n°0 and can be recalled with function "Load parameters in use in cell 0". The name of the loaded program will be shown in the bottom left corner of the normal use screen. |           |                            |                  |         |

### 3.6.3 MANUAL menu

| Parameter                                                                                                                                                                                                                                                                                           | Remote | Possible settings | Standard setting | Options                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CYL 1                                                                                                                                                                                                                                                                                               | -      | -                 | -                |   |
| CYL 2                                                                                                                                                                                                                                                                                               |        |                   |                  |                                                                                                                                                                             |
| This command, when enabled, allows to move manually the soldering wire cylinder.<br>To move it use  (backward) and  (forward) |        |                   |                  |                                                                                                                                                                             |

| Parameter                                                                                                                                                                                                                                                                                                         | Remote | Possible settings | Standard setting | Options                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CYL 1 and 2</b>                                                                                                                                                                                                                                                                                                | -      | -                 | -                |   |
| This command, when enabled, allows to move manually both soldering wire cylinders, together.<br>To move them use  (backward) and  (forward) |        |                   |                  |                                                                                                                                                                             |

| Parameter                                                                                                                                                                                                                                                                                        | Remote | Possible settings | Standard setting | Options                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WIRE 1</b>                                                                                                                                                                                                                                                                                    | -      | -                 | -                |   |
| <b>WIRE 2</b>                                                                                                                                                                                                                                                                                    |        |                   |                  |                                                                                                                                                                             |
| This command, when enabled, allows to feed manually the soldering wire.<br>To feed the wire use  (backward) and  (forward) |        |                   |                  |                                                                                                                                                                             |

## 4 MAINTENANCE



Before performing any maintenance operation on the unit, read carefully the *Safety instructions - Warnings* section of this manual.

### 4.1 Periodical maintenance

All periodical maintenance operations are listed below, together with the preferred frequency:

| Intervention | Frequency    | Operator | Procedure           |
|--------------|--------------|----------|---------------------|
| Cleaning     | Periodically | Operator | See paragraph 4.2.1 |



The operator shall evaluate the system status and its working capability, according eventually with the maintenance personnel.

### 4.2 Periodical maintenance procedures

#### 4.2.1 Cleaning

Do not wash the device with liquid detergents or chemical substances. Any cleaning should be done using a slightly-damp, non-abrasive cloth.

### 4.3 Troubleshooting

See paragraph 4.4 for the self-diagnosis messages list.

#### 4.4 Self-diagnosis messages

The microcomputer controller notifies the operator in case of malfunctioning of one of the devices connected to the Master Controller by means of a **self-diagnosis** display message.

The Master Controller display is divided, in this case, into two sections: the first line at the top displays the status of station 1, the second line displays the status of station 2 (see illustration below).

Also in cases where the self-diagnosis system gives a signal, the signal will appear in one of the two zones depending on whether it occurred in the first or second station.

The illustration beside shows an example of self-diagnosis that relates to station 1 (generator off).

```

1> A12: generator
    not ready
2> T= 700C S=====L
    TIME= 10s POW.=99%
  
```

| Message                                                                                                                                                                                                                                  | Remote message | Probable cause / Action                                                                                                                                    | Signal on Fieldbus |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| –                                                                                                                                                                                                                                        | R              | No diagnosis, ready to operate                                                                                                                             | 0                  |
| No actions required                                                                                                                                                                                                                      |                |                                                                                                                                                            |                    |
| –                                                                                                                                                                                                                                        | P              | No diagnosis, cycle running                                                                                                                                | 0                  |
| No actions required                                                                                                                                                                                                                      |                |                                                                                                                                                            |                    |
| A1: water temperature too high                                                                                                                                                                                                           | N [A1]         | Water temperature too high                                                                                                                                 | 1                  |
| Check the temperature of cooling water that have to be less than 50° (intervention threshold =55°C)                                                                                                                                      |                |                                                                                                                                                            |                    |
| A2: internal fault                                                                                                                                                                                                                       | N [A2]         | Tuning failure                                                                                                                                             | 2                  |
| Contact the technical department of CEIA or the nearest distributor                                                                                                                                                                      |                |                                                                                                                                                            |                    |
| A3: no water                                                                                                                                                                                                                             | N [A3]         | The flow of cooling water has stopped                                                                                                                      | 3                  |
| Check the presence of water in the cooling unit.                                                                                                                                                                                         |                |                                                                                                                                                            |                    |
| Contact the SUPERUSER to check the fluxometer input in the J4-J5 connectors.                                                                                                                                                             |                |                                                                                                                                                            |                    |
| Contact the SUPERUSER to check the presence of blocking-up in the generator inlets and outlets                                                                                                                                           |                |                                                                                                                                                            |                    |
| A4: increase coil dimension                                                                                                                                                                                                              | R [A4]         | Too small dimensions of the inductor.                                                                                                                      | 4                  |
| A4: add a loop                                                                                                                                                                                                                           |                |                                                                                                                                                            |                    |
| Contact the SUPERUSER to increase coil dimensions                                                                                                                                                                                        |                |                                                                                                                                                            |                    |
| A4: increase coil dimension                                                                                                                                                                                                              | R [A4]         | Serie 50 generators: wrong coupling between the inductor and the piece to heat (the message A4 is shown after 20s from the start of the heating workcycle) | 4                  |
| A4: add a loop                                                                                                                                                                                                                           |                |                                                                                                                                                            |                    |
| Contact the SUPERUSER to verify the correct dimensioning of the inductor through the free software PowerCube Assistant                                                                                                                   |                |                                                                                                                                                            |                    |
| A4: check short circuit in adductors                                                                                                                                                                                                     | R [A4]         | Short circuit in the inductor                                                                                                                              | 4                  |
| Check the absence of short circuit of lead in wires                                                                                                                                                                                      |                |                                                                                                                                                            |                    |
| A5: decrease coil dimensions                                                                                                                                                                                                             | R [A5]         | Coil too big                                                                                                                                               | 5                  |
| Contact the SUPERUSER to decrease coil dimensions. Eventually verify and optimize through the free software PowerCube Assistant                                                                                                          |                |                                                                                                                                                            |                    |
| A5: decrease coil dimensions                                                                                                                                                                                                             | R [A5]         | Rims too far                                                                                                                                               | 5                  |
| Contact the SUPERUSER to check that the distance between the rims of the head is not greater than the minimum required distance (see the Generator Manual). Eventually verify and optimize through the free software PowerCube Assistant |                |                                                                                                                                                            |                    |



| Message                                                                                                                                                                                                                                                                                                                                                                                                  | Remote message | Probable cause / Action                                                                                                                                           | Signal on Fieldbus |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| A5: decrease length of adductors                                                                                                                                                                                                                                                                                                                                                                         | R [A5]         | Rims too long                                                                                                                                                     | 5                  |
| Contact the SUPERUSER to diminish the length of the head spouts (centimetre by centimetre). The reduction of the length must be carried out <b>only in the event that the shape of the coil cannot be changed for the type of processing in progress and when the distance between the lead-in wires has been checked</b> . Eventually verify and optimize through the free software PowerCube Assistant |                |                                                                                                                                                                   |                    |
| A5: decrease length of adductors                                                                                                                                                                                                                                                                                                                                                                         | R [A5]         | <b>Serie 50 generators:</b> wrong coupling between the inductor and the piece to heat (the message A5 is shown after 30s from the start of the heating workcycle) | 5                  |
| Contact the SUPERUSER to verify the correct dimensioning of the inductor through the free software PowerCube Assistant                                                                                                                                                                                                                                                                                   |                |                                                                                                                                                                   |                    |
| A6: internal fault                                                                                                                                                                                                                                                                                                                                                                                       | R [A6]         | Possible breakdown of an internal component                                                                                                                       | 6                  |
| Contact the technical department of CEIA or the nearest distributor                                                                                                                                                                                                                                                                                                                                      |                |                                                                                                                                                                   |                    |
| A7: head not connected                                                                                                                                                                                                                                                                                                                                                                                   | R [A7]         | Disconnection of a head                                                                                                                                           | 7                  |
| Contact the SUPERUSER to check the connection between the heads and the electronic unit                                                                                                                                                                                                                                                                                                                  |                |                                                                                                                                                                   |                    |
| A8: generator timeout                                                                                                                                                                                                                                                                                                                                                                                    | R [A8]         | Generator timeout                                                                                                                                                 | 8                  |
| <b>Message coming only from Spectacle Frames Line generators:</b> The generator activation time is too long:<br>Contact the SUPERUSER to check the programming and reduce the activation time.                                                                                                                                                                                                           |                |                                                                                                                                                                   |                    |
| A9: no wire                                                                                                                                                                                                                                                                                                                                                                                              | R [A9]         | Soldering wire not present                                                                                                                                        | 8                  |
| There is no more soldering wire in the feeder. Load the feeder with wire.                                                                                                                                                                                                                                                                                                                                |                |                                                                                                                                                                   |                    |
| Contact the SUPERUSER to verify the correct positioning of the soldering wire in respect to the sensor.                                                                                                                                                                                                                                                                                                  |                |                                                                                                                                                                   |                    |
| If a sensor is not installed, contact the SUPERUSER to verify that the "Wire presence" parameter is set to OFF                                                                                                                                                                                                                                                                                           |                |                                                                                                                                                                   |                    |
| A10: low line voltage                                                                                                                                                                                                                                                                                                                                                                                    | N [A10]        | Generator power supply under the range                                                                                                                            | 10                 |
| Contact the SUPERUSER to insert (between the mains and the heater) a transformer with an input voltage equal to that available on the line and an output voltage correct for the connected Generator                                                                                                                                                                                                     |                |                                                                                                                                                                   |                    |
| A11: high line voltage                                                                                                                                                                                                                                                                                                                                                                                   | N [A11]        | Generator power supply over the range                                                                                                                             | 11                 |
| Contact the SUPERUSER to insert (between the mains and the heater) a transformer with an input voltage equal to that available on the line and an output voltage correct for the connected Generator                                                                                                                                                                                                     |                |                                                                                                                                                                   |                    |
| A12: generator not ready                                                                                                                                                                                                                                                                                                                                                                                 | N [A12]        | PowerCube not connected                                                                                                                                           | 12                 |
| Check that the generators are switched ON.<br>Contact the SUPERUSER to check the connection of Power Cubes and their functioning                                                                                                                                                                                                                                                                         |                |                                                                                                                                                                   |                    |
| A13: gas timeout                                                                                                                                                                                                                                                                                                                                                                                         | R [A13]        | Gas timeout (internal timer)                                                                                                                                      | 13                 |
| Contact the SUPERUSER to check the proper operating of external devices.                                                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                                   |                    |
| Contact the SUPERUSER to check the functioning parameters.                                                                                                                                                                                                                                                                                                                                               |                |                                                                                                                                                                   |                    |
| A14: heating timeout                                                                                                                                                                                                                                                                                                                                                                                     | N [A14]        | Heating timeout (internal timer)                                                                                                                                  | 14                 |
| Contact the SUPERUSER to check the proper operating of external devices.                                                                                                                                                                                                                                                                                                                                 |                |                                                                                                                                                                   |                    |
| Contact the SUPERUSER to check the functioning parameters.                                                                                                                                                                                                                                                                                                                                               |                |                                                                                                                                                                   |                    |
| A15: wire timeout                                                                                                                                                                                                                                                                                                                                                                                        | R [A15]        | Wire feeding timeout                                                                                                                                              | 15                 |
| Contact the SUPERUSER to check the proper operating of the soldering wire feeder.                                                                                                                                                                                                                                                                                                                        |                |                                                                                                                                                                   |                    |
| Contact the SUPERUSER to check the parameters programming                                                                                                                                                                                                                                                                                                                                                |                |                                                                                                                                                                   |                    |
| A16: phase missing                                                                                                                                                                                                                                                                                                                                                                                       | N [A16]        | Phase missing                                                                                                                                                     | 16                 |
| Message coming from tri-phase generators only.<br>Contact the SUPERUSER to check the presence of all of the three phases in the main.                                                                                                                                                                                                                                                                    |                |                                                                                                                                                                   |                    |

| Message                                                                                                                                                                                                                                                                                                                                       | Remote message | Probable cause / Action            | Signal on Fieldbus |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------|--------------------|
| A101:temp.sensor 1 not found                                                                                                                                                                                                                                                                                                                  | N[A101]        | Temperature sensor 1 not connected | 101                |
| Contact the SUPERUSER to check the connections and the temperature sensor status.<br>This diagnosis has to be activated with P1DIAG ON                                                                                                                                                                                                        |                |                                    |                    |
| A102:temp.sensor 2 not found                                                                                                                                                                                                                                                                                                                  | N[A102]        | Temperature sensor 2 not connected | 102                |
| Contact the SUPERUSER to check the connections and the temperature sensor status.<br>This diagnosis has to be activated with P2DIAG ON                                                                                                                                                                                                        |                |                                    |                    |
| A121: PLC COMM                                                                                                                                                                                                                                                                                                                                | R[A121]        | Missing PLC communication          | 121                |
| Contact the SUPERUSER to check the PLC connections and programming.<br>With connections re-established and programming verified, start a new cycle to reset the diagnosis                                                                                                                                                                     |                |                                    |                    |
| UPP sen. not found                                                                                                                                                                                                                                                                                                                            | -              | UPP® sensor not found              | -                  |
| Contact the SUPERUSER to check the sensor power supply, its serial connection and in the end, the parameters.<br>On the Controller, verify that the hidden parameter MODE > output supply is set on 24V.<br>On UPP sensor check the following parameters: baud rate = 57600; address = 00; wait time = 0; switch for serial interface = RS232 |                |                                    |                    |
| UPP sensor not ready                                                                                                                                                                                                                                                                                                                          | -              | UPP® sensor not ready              | -                  |
| Wait until the sensor ends the self-balancing procedure.                                                                                                                                                                                                                                                                                      |                |                                    |                    |
| UPP sensor internal fault                                                                                                                                                                                                                                                                                                                     | -              | UPP® sensor faulty                 | -                  |
| Contact the Assistance service for the sensor.                                                                                                                                                                                                                                                                                                |                |                                    |                    |
| -                                                                                                                                                                                                                                                                                                                                             | -              | Temperature 1 out of tolerance     | 111                |
| Contact the SUPERUSER to check the programming parameters.                                                                                                                                                                                                                                                                                    |                |                                    |                    |
| -                                                                                                                                                                                                                                                                                                                                             | -              | Temperature 2 out of tolerance     | 112                |
| Contact the SUPERUSER to check the programming parameters.                                                                                                                                                                                                                                                                                    |                |                                    |                    |

As soon as the malfunctioning occurs, the apparatus notifies the operator by means of a short alarm signal.

#### 4.4.1 Error messages during use of the memory cells

| Message                                               | Cause                                                                                   | Possible intervention                                                                                                                                                                                                     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NO PARAMETERS SET FOUND<br>Press any key              | Attempt to load a set of parameters from an empty memory cell                           | Press the and key to give confirmation and repeat the loading operation using a correctly programmed memory                                                                                                               |
| NOT VALID OR DEFECTIVE CARD<br>Extract the card       | Attempt to load a set of parameters from a defective or not valid Program Card          | Extract the card and repeat the action with a valid Program Card.                                                                                                                                                         |
| PARAMETERS SET CORRUPTED<br>E = continue<br>P = abort | Attempt to load a set of parameters from a memory cell, that was not correctly recorded | It is possible to interrupt the loading by pressing the <b>P</b> key or to continue by pressing <b>E</b> . In cases where the loading is carried out, it is advisable to check the parameters before using the apparatus. |